Proposal for grant giovani in CSN5 - Year 2026

Acronym:	PRISM
Full title:	Precision Room-temperature Instrumentation for high-energy X-ray absorption Spectroscopy Measurements
Applicant:	Simone Manti
Participating units:	INFN National Laboratory of Frascati

Abstract

X-ray absorption spectroscopy (XAS) in the laboratory above 20 keV remains a challenge in hard X-ray instrumentation. Although synchrotron facilities provide outstanding performance, their limited accessibility motivates laboratory instruments for routine characterization. In this context, the limited photon flux of laboratory X-ray sources for fluorescence-mode XAS on thick samples makes high detection efficiency above 20 keV essential. However, silicon detectors lose efficiency at hard X-ray energies, while high-purity germanium detectors require cryogenic operation. Therefore, access to K-edges above 20 keV in the laboratory for elements such as Mo, Pd, Ag and Sn remains limited, despite their importance in catalysis, electrochemistry, and energy materials. PRISM will establish a laboratory platform for quantitative fluorescence-mode XAS above 20 keV by combining high-resolution crystal optics with a room-temperature multichannel cadmium zinc telluride (CZT) detector. PRISM integrates crystal monochromatization, energy-resolving detection, dedicated electronics, waveform reconstruction and SHADOW4-GEANT4 optimization into a single instrument. Building on expertise gained from the VOXES project and SIDDHARTA experiment, PRISM will develop the complete detection chain, targeting for the CZT an energy resolution ≤2% at 22.2 keV with high efficiency and SNR. The platform will be validated through fluorescence XANES at the Ag K-edge and demonstrated on realistic thick and heterogeneous samples. Beyond the proof of concept, PRISM will provide INFN with new capabilities in hard X-ray spectroscopy and semiconductor detector instrumentation. Its CZT detector, algorithms, calibration procedures and simulation framework will provide reusable technologies for future X-ray instruments. By bridging laboratory spectroscopy and large-scale facilities, PRISM will enable flexible, high-performance hard X-ray spectroscopy currently inaccessible outside synchrotron beamlines.

1 State of the art

A central challenge in radiation-detector development is enabling **in-situ and operando characterization under realistic conditions**. This requires element- and bulk-sensitive measurements of weak signals from thick, heterogeneous or dilute samples over long acquisition campaigns [1,2].

X-rays are particularly suited for such investigations because of their high penetration depth, enabling access to the bulk of samples [3]. Among X-ray techniques, **X-ray absorption spectroscopy (XAS)** directly probes absorption across an element-specific absorption edge [4]. The near-edge region (XANES) is sensitive to oxidation state, electronic structure and coordination geometry, whereas the extended region up to 1 keV above the edge (EXAFS) provides quantitative information on the local atomic environment. XAS can be measured in transmission, from beam attenuation through the sample, or in fluorescence, from the emitted characteristic X-rays. **Fluorescence detection is particularly suited to dilute, thick or geometrically constrained samples** for which transmission is impractical [5].

Synchrotron facilities are the reference platforms for XAS, as their high brilliance, energy tunability and dedicated beamlines enable high-quality spectra with excellent SNR over a wide energy range [6]. However, **access to these large-scale facilities is limited and competitive**, and the allocated beamtime is often unsuitable for routine characterization, long experimental campaigns, or studies involving hazardous and radioactive samples [7]. These limitations have driven renewed interest in **laboratory XAS systems based on conventional X-ray sources** [8]. Although laboratory sources provide lower brilliance than synchrotrons, they offer continuous access, flexible sample handling and long measurement times, making them attractive for routine analysis, method development and slowly evolving in-situ or operando processes [9].

Recent advances in source brightness, monochromator crystals and detection technology have enabled **an increasing number of XAS and XES measurements to be transferred back to the laboratory** [10,11]. Commercial laboratory XAS systems are available (e.g. Rigaku, Sigray, easyXAFS), confirming the maturity and industrial relevance of the field, although their architecture and substantial cost limit flexibility for customized fluorescence measurements. Within INFN-LNF, **the VOXES spectrometer** [12,13] has established expertise in high-resolution von Hamos spectroscopy for extended and dilute X-ray sources. The instrument currently operates mainly in the 5–20 keV range and has recently been upgraded with motorized positioning, energy-dispersive monitoring and mechanical compatibility with transmission XAS measurements [14].

Despite this progress, **laboratory fluorescence-XAS above approximately 20 keV remains challenging** [15], although this range includes many technologically relevant K-edges of Mo, Pd, Ag and Sn, whose lower-energy L-edges are too closely spaced [16]. Silicon-based detectors progressively lose absorption efficiency in the hard-X-ray range, whereas high-purity germanium detectors offer excellent spectroscopic performance but require cryogenic cooling. **High-Z compound semiconductors such as CdTe and CdZnTe instead provide high hard-X-ray absorption efficiency and room-temperature operation**. Although CZT detectors have been developed for synchrotron, imaging and high-flux applications, **their integration into multichannel, energy-resolving laboratory fluorescence-XAS systems remains largely unexplored** [17,18].

PRISM is specifically designed to close this technological gap by combining three mature but not previously integrated components into a single optimized platform: integrating segmented CZT sensors, low-noise electronics with waveform-based reconstruction and high-resolution crystal optics. Such a platform would extend laboratory fluorescence-XAS beyond 20 keV while **consolidating the INFN expertise in hard-X-ray detection and spectroscopy.**

2 Goals and Expected Outcome

The **main goal of the project is to develop and validate a room-temperature laboratory platform for fluorescence-mode XAS above 20 keV, combining a high-resolution crystal monochromator with a custom multichannel CZT detector**. The X-ray source, crystal optics, sample environment, detector, electronics, data acquisition, and simulation framework will be optimized as an integrated measurement chain for quantitative XANES studies of samples unsuitable for transmission measurements because of thickness, dilution, or geometry. **To my knowledge, no existing laboratory system combines crystal monochromatization with multichannel energy-resolving CZT detection for quantitative fluorescence XAS above 20 keV**.

The project is structured around **three goals (G)** and corresponding **expected outcomes (EO)**:

G1: Design, construct and validate the complete fluorescence-mode PRISM XAS instrument through end-to-end simulations and experimental commissioning.
EO1: A fully commissioned laboratory setup with optimized source–optics–sample–detector geometry, supported by a validated simulation framework. Simulated and measured count rates will agree within 20%, while beam spot size and incident-energy resolution will be validated within 10%.

G2: Develop and characterize a room-temperature multichannel CZT detection system optimized for hard X-ray fluorescence spectroscopy.
EO2: A calibrated CZT detector achieving intrinsic detection efficiency above 90% at 50 keV, energy resolution at the ≤2% level for the Ag Kα fluorescence line at 22.5 keV, and dead time below 10% under the operating conditions selected for XANES acquisition.

G3: Demonstrate PRISM's quantitative fluorescence XANES measurements above 20 keV using the Ag K-edge as benchmark.
EO3: Ag K-edge XANES spectra will be acquired for metallic Ag foil and two Ag(I) reference compounds with different local coordination environments. The measurements will achieve a monochromatic incident-beam bandwidth of ≤10 eV FWHM and edge-step SNR ≥ 20.

3 Methodology

XAS in the laboratory

XAS measures the energy dependence of the linear absorption coefficient, μ(E), across an element-specific core-level absorption edge **(see Figure 1)**. The **near-edge region (XANES) is sensitive to oxidation state, orbital occupancy and local symmetry**, whereas the oscillations **above the edge (EXAFS) provide information on coordination numbers, interatomic distances and structural disorder**. Laboratory XAS becomes increasingly demanding at hard-X-ray energies because the usable flux from conventional sources decreases and EXAFS requires accurate measurements over a broad energy interval. **PRISM will use the Ag K-edge at 25.5 keV as its benchmark, first establishing a robust XANES protocol and then quantifying the EXAFS range achievable within laboratory-compatible acquisition times.**

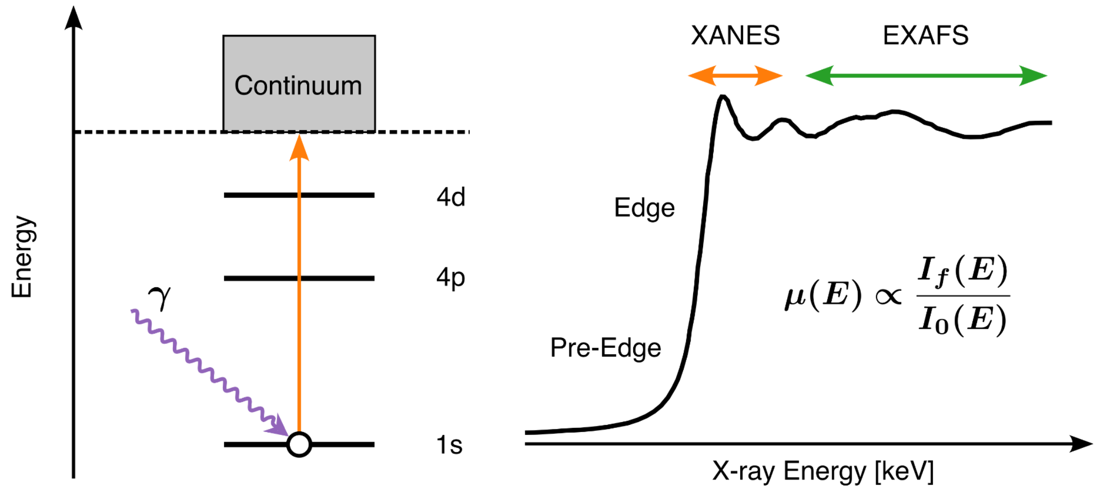

FIGURE 1. Principles of XAS illustrated for the Ag K-edge. Left: schematic of the electronic transitions involved in X-ray absorption. When the incident photon energy exceeds the Ag K-edge (≈25.5 keV), a 1s core electron is excited into the continuum, creating a core hole. Right: qualitative evolution of the absorption coefficient, μ(E), as a function of photon energy, highlighting the XANES (X-ray Absorption Near Edge Structure) and EXAFS (Extended X-ray Absorption Fine Structure) regions. XANES is primarily sensitive to the oxidation state and local electronic structure, while EXAFS provides quantitative information on the local atomic environment, including coordination numbers and interatomic distances.

The absorption coefficient μ(E) can be acquired in transmission or fluorescence mode. In a dispersive transmission arrangement, such as that being explored with VOXES [14], the broad source spectrum passes through the sample and a monochromator crystal spatially separates the transmitted energies onto a position-sensitive detector. Comparison of the spectra acquired with (I(E)) and without (I_0(E)) the sample provides μ(E). This approach is efficient but requires a homogeneous sample with an appropriate transmission thickness. **PRISM will instead operate primarily in fluorescence mode**. A crystal placed before the sample will select and focus one incident energy at a time, while an energy-resolving detector positioned approximately perpendicular to the incident beam will measure the characteristic fluorescence emitted following core-hole creation. For the Ag K-edge (see Figure 1), I_f(E) will be extracted from the Ag Kα line complex at approximately 22.2 keV. This geometry extends XAS to thick, dilute, heterogeneous, or geometrically constrained samples that are impractical in transmission. In the thin or dilute limit, after background, live-time and detector-response corrections, the absorption spectrum is proportional to the ratio $\mu(E) \propto I_f(E) / I_0(E)$ where I_0(E) is the incident monochromatic intensity.

PRISM will extend fluorescence-mode XAS above 20 keV by combining high-order crystal optics with efficient room-temperature CZT detection. A cylindrically bent germanium crystal, using the Ge(880) reflection as the baseline configuration, provides a sufficiently small lattice-plane spacing to satisfy the Bragg condition at practical angles and achieve the energy dispersion required in this range. The incident energy will be selected from the X-ray-tube continuum in a VOXES-derived von Hamos geometry, while the CZT array will maintain high detection efficiency for the corresponding fluorescence photons. During an energy scan, the crystal rotation will be coordinated with the motorized axes to preserve the source–crystal–sample geometry and the position of the focused spot. At each energy set point, a complete fluorescence spectrum will be acquired for a defined dwell time, and $I_f(E)$ will be extracted by fitting or integrating the selected fluorescence line after local-background subtraction. Finer energy steps will be used around the absorption edge and coarser steps in the extended region, with dwell times adapted to the predicted count rate and target statistical uncertainty.

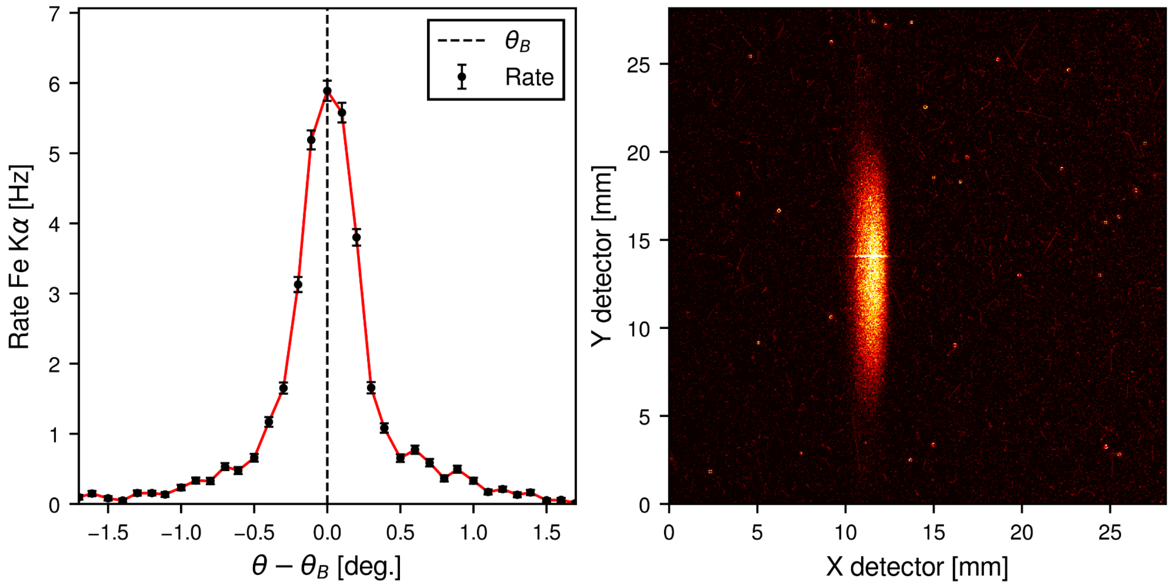

FIGURE 2. Performance of the VOXES setup in angular scanning and X-ray focusing. Left: angular dependence of the Fe Kα count rate measured using a mosaic graphite (002) crystal, showing a reflection profile extending over approximately 1°. Right: millimetre-scale Fe Kα focal spot recorded with a Timepix detector featuring a 28 x 28 mm^2 active area.

The feasibility of the required angular scan and focusing is supported by the existing VOXES experience. For instance, Figure 2 (left) shows the measured Fe Kα rate on the position-sensitive detector as the crystal is rotated through its nominal Bragg angle, demonstrating a controlled Bragg-angle scan. In Figure 2 (right), the same Fe Kα line is shown on a two-dimensional detector, exhibiting millimetric size spot. Although these measurements were performed in an X-ray-emission configuration and do not themselves constitute XAS, they validate two essential instrumental functions for PRISM: **precise crystal rotation for energy selection and spatial focusing with a bent crystal**. The project will build on this expertise, the available VOXES components and in-kind contributions, specifically the two slits to collimate the flux after the source, existing crystal rotation and translation stages, and the shielded laboratory infrastructure at LNF.

CZT detector and signal reconstruction

The detector represents the key technological bottleneck for laboratory hard-X-ray fluorescence XAS. **PRISM will overcome this limitation by developing a custom multichannel CZT detector optimized for laboratory fluorescence measurements in the 20–50 keV range**. The design will build on the experience gained within SIDDHARTA, in collaboration with IMEM-CNR in Parma and with the University of Palermo, where CZT detectors were tested at the DAΦNE collider under high-flux conditions [19–22].

The baseline CZT detector will consist of **a monolithic 2-mm-thick CZT crystal segmented into a 20-pixel linear array with a 500 µm pitch**, common cathode and surrounding guard ring [23]. This geometry will increase the active area and collected solid angle, while distributing the fluorescence rate among independent channels to reduce pile-up and enable channel-resolved background monitoring. Each pixel will be coupled to a dedicated low-noise preamplifier. The system will be characterized in terms of efficiency, energy resolution, gain linearity and stability, dead time, rate capability and channel-to-channel uniformity.

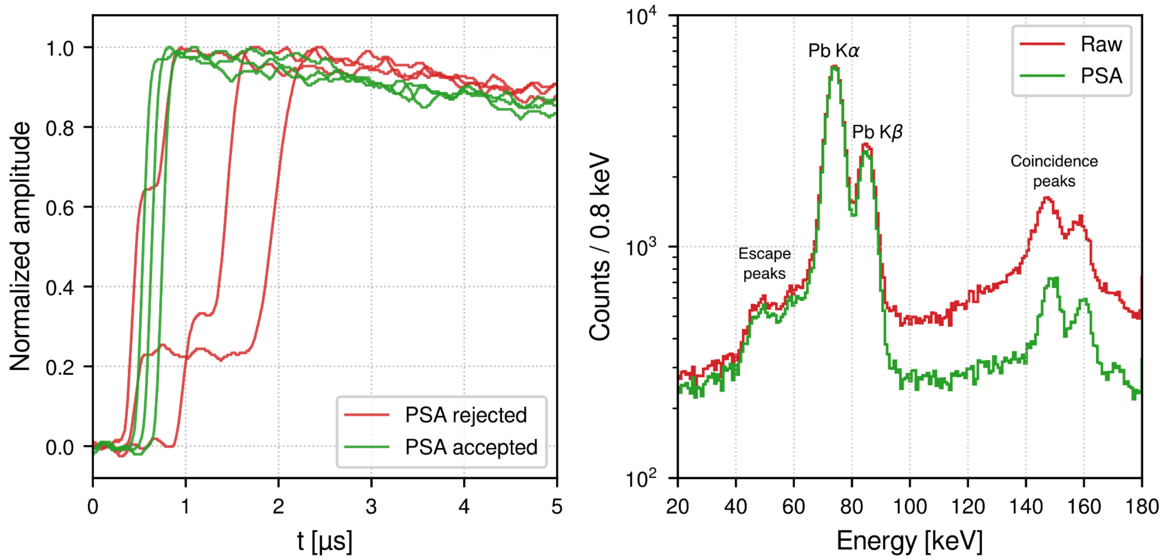

FIGURE 3. Machine-learning-assisted pulse-shape analysis (PSA) of CZT detector signals. Left: representative normalized waveforms accepted by the PSA as clean single-step events (green) and rejected multi-step events (red). Right: calibrated energy spectrum before (Raw) and after PSA selection, showing the preservation of the Pb Kα and Kβ fluorescence peaks and the suppression of the continuum, escape and coincidence structures. Results presented at the XGRADE 2026 workshop [23].

Waveforms will be digitized using a 64-channel, 16-bit, 125 MS/s acquisition platform. Although the detector requires 20 primary channels, the larger channel capacity will accommodate guard-ring, synchronization and diagnostic signals, provide margin for future detector expansion. A detector-response simulation will couple GEANT4 energy-deposition calculations to electric-field and weighting-potential models. Charge transport will then be simulated to generate channel-resolved current pulses including trapping, incomplete charge collection and charge sharing between adjacent pixels. After convolution with the measured front-end response and addition of realistic electronic noise, the synthetic waveforms will be used to develop and validate the reconstruction algorithms. Conventional pulse-shape observables and machine-learning classifiers, building on experience with germanium detectors **[24] and ongoing CZT tests at the LNF Beam Test Facility [25] (see Figure 3)**, will distinguish genuine fluorescence events from pile-up, saturation, electronic disturbances and distorted pulses. Dedicated reconstruction algorithms will also identify or correct escape-peak contributions, charge-sharing events and incomplete charge collection, thereby improving fluorescence-peak reconstruction and spectral quality.

The same CZT detector head will measure both the fluorescence signal $I_f(E)$ and the incident intensity $I_0(E)$ through a motorized exchange system. The CZT head and sample holder will each be mounted on independent rotation stages, while a long-travel linear axis will alternate the sample and detector between the two configurations. For $I_f(E)$, the sample will be placed at the monochromatic focus and observed by the CZT array at approximately 90°; for $I_0(E)$, the sample will be translated out and the detector rotated and repositioned into the attenuated direct beam. Calibrated attenuators, shorter dwell times and, when required, reduced tube current or slit aperture will keep the direct-beam rate within the validated linear operating range.

Source to sample simulation

To identify the optimal instrument configuration within the available project budget, **PRISM will develop an end-to-end, source-to-detector simulation of the entire spectrometer**. The Ag K-edge at approximately 25.5 keV will be used as the reference case for defining the XANES measurement protocol and the required performance in terms of energy resolution, photon flux, SNR and acquisition time. The simulation workflow will consist of two coupled stages. **First, SHADOW4 [26] ray-tracing simulations building on the approach previously validated for the VOXES spectrometer [27]**, which will model photon transport from the X-ray tube through the collimating slits and crystal optics to the sample.

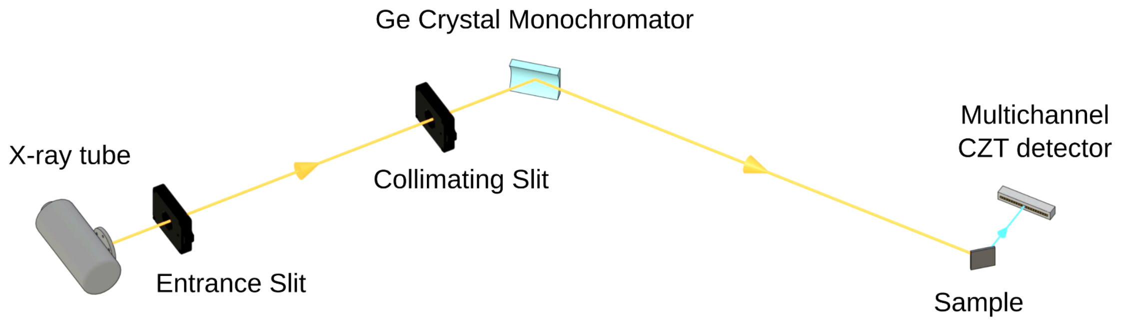

FIGURE 4. Conceptual layout of the proposed PRISM laboratory fluorescence-XAS setup. Radiation from the X-ray tube is collimated by two adjustable slits and monochromatized by a sagittally curved germanium crystal. The diffracted beam illuminates the Ag sample, while the emitted fluorescence is collected off-axis by a linear multichannel CZT detector. The drawing is not to scale, and slit and detector positions are illustrative.

Different germanium-based crystal reflections will be investigated in the VOXES-type von Hamos geometry, with the Ge(880) considered as the baseline candidate for measurements around the Ag K-edge [15]. The contributions from lower harmonics (e.g. (220), (440), and (660)) will be taken into account. Parametric studies will optimize the crystal material, reflection, curvature radius, dimensions, source–crystal and crystal–sample distances, slit apertures and Bragg geometry. The simulations will quantify the geometrical acceptance, energy bandwidth, photon flux, beam profile and illuminated spot size at the sample, while also evaluating alignment and mechanical tolerances. **The SHADOW4 output will then be transferred to GEANT4, which will simulate photon absorption in the sample, fluorescence emission, self-absorption, background contributions and energy deposition in the multichannel CZT detector**. The combined simulations will be used in the first months of the project to optimize the source, crystal optics, slit system, sample geometry and detector configuration, balancing count rate, spectral resolution and background rejection. A coordinated motorized scanning scheme will also be defined to vary the incident energy across the Ag K-edge while maintaining the required source–crystal–sample geometry. The final outcome of this activity will be a design freeze specifying the X-ray source, crystal type and dimensions, curvature radius, distances, slit apertures, sample position, CZT geometry, motion stages and required positioning accuracy. **A proposed layout of the setup for PRISM is shown in Figure 4, highlighting (not to scale) all the components from the source, the slit system, crystal, sample and the custom CZT detector.**

4 Organisation

Name	Unit	FTE	Responsibility/WP
Simone Manti	INFN, National Laboratory of Frascati	1.0	Project Leader; WP1–WP4
Alessandro Scordo	INFN, National Laboratory of Frascati	0.3	Crystal optics, ray tracing, integration; WP1/WP3
Massimiliano Bazzi	INFN, National Laboratory of Frascati	0.3	Electronics and DAQ; WP2
Catalina Curceanu	INFN, National Laboratory of Frascati	0.1	Scientific Supervisor and application validation; WP4
Total		**1.7**	**-**

PRISM will be divided into the following Work Packages (WP):

- **WP1 – Requirements, simulations and design freeze (M1–M6):** The experimental requirements will be established through an end-to-end SHADOW4–GEANT4 simulation workflow describing the complete measurement chain, from X-ray generation and crystal optics to sample response and CZT detection. The simulations will define the optimized source–optics–sample–detector geometry, performance targets and procurement specifications, minimizing technical risks before construction. Dr. Alessandro Scordo will contribute his expertise in crystal optics, ray tracing and VOXES-based spectrometer design.
- **WP2 – Room-temperature multichannel CZT spectroscopy for hard-X-ray fluorescence XAS (M4–M15):** A multichannel CZT detector with dedicated low-noise front-end electronics and waveform-based DAQ will be developed and characterized for hard-X-ray fluorescence spectroscopy. Channel calibration, gain equalization, efficiency measurements and evaluation of energy resolution, stability, dead time and pile-up performance will be performed. Pulse-shape-analysis methods will be implemented to maximize spectral quality and identify distorted events. Ing. Massimiliano Bazzi will support the development and validation of the electronics and DAQ chain.
- **WP3 – Instrument integration and commissioning (M10–M18):** After the design freeze, the X-ray source, crystal optics, motion stages, slits and mechanical components will be procured and validated through acceptance tests. The complete system will then be mechanically and electronically integrated. Commissioning will include optical alignment, synchronized energy scanning, control-system validation and quantitative comparison of measured flux, spot size and energy resolution with simulation predictions.
- **WP4 – Ag reference samples and realistic sample demonstration (M16–M24):** The complete instrument will first be validated using Ag foil and reference Ag compounds with different chemical states. Quantitative XANES performance will be assessed through energy

calibration, reproducibility, signal-to-noise ratio and comparison with reference spectra. Fluorescence-XAS will then be demonstrated on a realistic thick, diluted or heterogeneous sample, defining the accessible measurement domain. I shall perform crystal-optics optimization, measurements and interpretation of the experimental results, with support from team members.

The project **Milestones (MS)** are:

- **MS1 – Instrument concept validated and design frozen (M6):** Optimized source–optics–sample–detector configuration established, with predicted flux, energy resolution, spot size and acquisition performance meeting the XANES requirements.
- **MS2 – CZT spectroscopic performance demonstrated (M15):** The detector achieves >90% efficiency at 50 keV, ≤2% energy resolution at 25.5 keV, dead time <10%, and pile-up contamination <5% under XAS operating conditions.
- **MS3 – Integrated instrument commissioned (M18):** Synchronized energy scanning and calibration operational; measured count rates within 20%, and spot size and energy resolution within 10% of simulation predictions.
- **MS4 – Fluorescence-XAS capability demonstrated (M24):** Ag K-edge XANES measurements achieved on the reference metallic foil with SNR ≥20; discrimination of two chemical states demonstrated in a realistic sample with statistically significant discrimination.

The **Work Packages** and the **Milestones** are summarized in the **Gantt table in Figure 5**.

The project **Deliverables (D)** are:

- **D1** – Validated SHADOW4–GEANT4 workflow, optimized instrument design and technical specifications (M6).
- **D2** – CZT calibration and performance-characterization dataset with validation report (M15).
- **D3** – Ag foil and reference-compound XANES dataset with quantitative validation report (M21).
- **D4** – Realistic-sample dataset, operating-domain assessment, documented analysis tools and final project report demonstrating PRISM capabilities (M24).

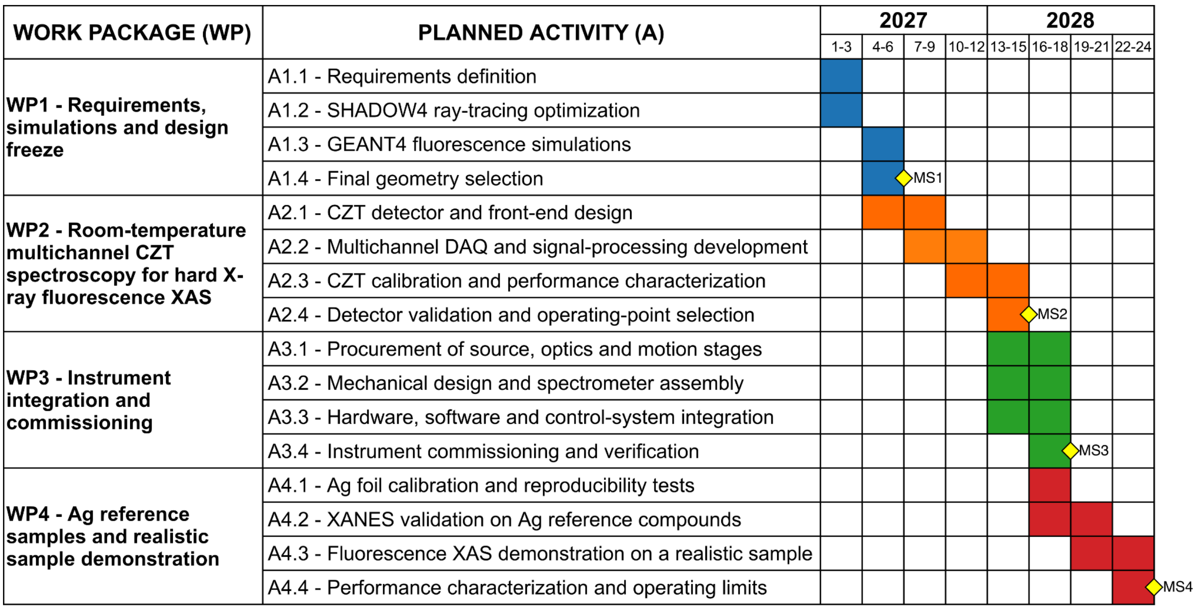

FIGURE 5. Gantt table of PRISM divided in each Work Packages (WPs) and for each one its Planned Activity (A) through the two years. Expected Milestones (MS) are indicated by the yellow marks.

5 Synergies with Other Activities

The PRISM project will establish extensive synergies with various external projects, institutions and companies.

Other INFN Scientific Committees

- **CSN3:** PRISM has synergy with KAONNIS, building on the experience gained with CZT detectors in the SIDDHARTA experiment at DAΦNE, and the VIP experiment at LNGS, which faces similar challenges in the detection of weak X-ray fluorescence lines above 20 keV [28] and employs related pulse shape analysis techniques [24].
- **CSN5:** Synergy with VOXES, a former CSN5 Young Researchers project whose expertise in crystal-based X-ray spectroscopy, instrumentation and data analysis provides the technological foundation for PRISM. VOXES will contribute existing equipment and laboratory infrastructure in kind, including facilities for testing and calibration with laboratory X-ray sources. Further synergies are foreseen with SPHINX, particularly in precision angular scanning, optical alignment and the integration of X-ray optics and detectors.
- **EuPRAXIA:** Synergy with EuAPS and the AQUA beamline, both targeting XAS measurements in different photon-energy ranges. Developing laboratory-based XAS methods within PRISM will provide complementary expertise, instrumentation and procedures useful for preparing and optimizing future experiments at these facilities.

External institutions and national and international research laboratories

- **University of Palermo:** PRISM will benefit from the established collaboration with the University of Palermo in the development and testing of CZT detectors within the

SIDDHARTA experiment. This synergy will support the optimization of the detector geometry, front-end electronics, calibration procedures and signal-reconstruction methods.

- **ENRICH COST Action:** PRISM is strongly aligned with the ENRICH COST Action, which promotes a European network for radiation-detection research and its applications to societal challenges. Within this network, the applicant is co-leader of the Working Group on Simulation. A particularly relevant connection is with Manuel Sánchez del Río, leader of the Working Group and principal developer of SHADOW4, the ray-tracing software that will be used to optimize the crystal optics and spectrometer geometry.

Industry and Private Sector

The following companies have provided letters of support for PRISM, confirming industrial interest in the proposed instrumentation and its technology-transfer potential.

- **Due2Lab:** Due2Lab has expressed interest in the PRISM project, particularly in view of its expertise in CZT crystal production and detector integration and the potential scalability of the proposed multichannel system.
- **Rigaku:** Rigaku has expressed interest in PRISM as a laboratory XAS development and may represent a relevant industrial reference for the selection of X-ray sources, optics, and other key components.

6 Impact and Spin-offs

For the INFN community, the impact will extend beyond the realization of a single instrument. **PRISM will valorize the existing VOXES infrastructure and consolidate LNF expertise in semiconductor radiation detectors, precision X-ray spectroscopy, radiation transport and spectral reconstruction**. The modular CZT detector and readout chain, validated source-to-detector simulations, calibration procedures and performance maps will provide reusable technological assets for INFN activities developing hard-X-ray instrumentation. The project will establish a transferable methodology linking predictive simulations, subsystem validation and quantitative performance assessment of integrated spectrometers. After laboratory validation, the resulting design principles may support future developments such as EuAPS. PRISM will also train young researchers and technical staff in optics, detectors, electronics, simulation and data analysis.

For the wider scientific community, **PRISM will extend laboratory access to fluorescence-XAS at the K-edges of technologically relevant elements such as Mo, Pd, Ag and Sn.** If the targeted performance is achieved, the platform will enable sample screening, method development, cell optimization and long-term in-situ or operando studies of thick, dilute and heterogeneous specimens that are difficult to investigate in transmission. PRISM will complement, rather than replace, synchrotron and XFEL facilities: preliminary characterization and protocol optimization will improve the efficiency of scarce beamtime, while repeated measurements and slowly evolving processes can be performed in the laboratory. Beneficiaries will include catalysis, electrochemistry, energy materials and advanced alloys. Documented workflows, reference datasets and operating-domain maps will support reproducibility and facilitate adoption by other laboratories.

The main technology-transfer opportunities are the modular room-temperature CZT detector, front-end and multichannel DAQ chain, together with the associated calibration and pulse-reconstruction methods, validated source–optics–sample–detector workflows and compact hard-X-ray spectrometer architecture. Following experimental validation, these assets may be evaluated with INFN technology-transfer structures and X-ray instrumentation companies for protection, further development or co-design. Potential applications include materials characterization, quality control and long-duration monitoring of catalysts, batteries and advanced materials.

7 Costs

The overall cost of **PRISM** is **112 k€**, where **66 k€** are requested for **2027** and **46 k€** for **2028**. The detailed costs are presented in Table I.

Year	Category	Item	Purpose/WP	Cost [€]
2027	Equipment	CZT Detector and Front-End	Fluorescence Detection - WP2	€30.000
2027	Equipment	Digitizer and DAQ	Signal Acquisition - WP2	€20.000
2027	Equipment	Motorized Stages and Controllers	Automated Positioning - WP3	€10.000
2027	Consumables	Ag Reference Targets	Calibration Tests - WP2/WP4	€2.000
2027	Services	Custom Mechanical Supports	Instrument Integration - WP3	€2.000
2027	Computing	DataCloud CPU and Storage	End-to-End Simulations - WP1	€2.000
Total 2027				**€66.000**
2028	Equipment	Cooled X-ray Tube	Excitation Source - WP3	€10.000
2028	Equipment	X-ray Tube Power Supply	Source Operation - WP3	€10.000
2028	Travel	Oral Conference Travel	Results Dissemination - WP4	€4.000
2028	Equipment	Bent Crystal Optics	Energy Selection - WP3	€20.000
2028	Consumables	Additional Reference Targets	Multi-Edge Validation - WP4	€2.000
Total 2028				**€46.000**
Total				**€112.000**

TABLE I. Detailed financial request for the two-year PRISM project, broken down by year, cost category, item, and related Work Package.

8 Risk Assessment

Here I will outline the possible **risks (R)** in the project in relation to the specific planned **activity (A)** and describe the possible mitigation.

R1. Delayed delivery of critical components (A3.1–A3.4)

- **Risk:** Delays in delivery of crystal optics, mechanics or detector components may affect the integration and commissioning schedule.
- **Mitigation:** Preliminary integration and alignment activities will be performed using compatible in-kind VOXES components, including non-optimized crystals, allowing system validation to proceed while final components are delivered.

R2. Extended CZT optimization phase (A2.1–A2.4)

- **Risk:** Calibration, characterization and optimization of the CZT detector and electronics may require additional time, delaying the XAS validation phase.
- **Mitigation:** An in-kind VOXES Si PIN diode will be used for preliminary calibration and reference measurements with reduced efficiency, while CZT optimization and performance improvements proceed in parallel.

R3. Limited accuracy of end-to-end simulations (A1.2–A1.4)

- **Risk:** SHADOW4–GEANT4 simulations may not fully reproduce the coupled response of source, crystal optics, sample and detector, affecting geometry optimization.
- **Mitigation:** The simulation workflow will be benchmarked against analytical calculations and existing VOXES measurements. Additional support will be provided through direct interaction with the SHADOW4 developer Manuel Sánchez del Río within COST Action ENRICH WG3.

R4. Longer-than-expected XANES acquisition time (A3.3, A4.1–A4.3)

- **Risk:** Limited photon flux or signal-to-background ratio may increase the acquisition time required for quantitative XANES measurements.
- **Mitigation:** Scan strategy, energy steps and dwell times will be optimized using simulation and commissioning data. If required, optical acceptance and detector shaping parameters will be adjusted while preserving the required energy resolution and background rejection performance.

9 References

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